

Building a Multi-Tenant Canonical Data Warehouse with Apache Spark

Ingestion Pipeline & Star Schema Modeling for Fragmented
Client Data

Problem Context & Schema Fragmentation

- **The Challenge:** Ingesting distinct text streams (CSV, Pipe, TSV) from multiple isolated clients representing identical business concepts (Customers and Transactions).
- **Identified Latent Landmines:**
 - **Schema Deviations:** Individual column mappings varied drastically (e.g., id vs customer_id vs cust_id vs user_id).
 - **Structural Variance:** Name structures were delivered split (Client A/C) vs completely collapsed into a single string column (Client B).
 - **Missing Dimensions:** Crucial temporal attributes were entirely omitted by certain sources (Client C Transactions).
 - **Data Formatting Corruptions:** Currency annotations embedded raw symbols (\$) and inline logging artifacts ('EXTRA') threatened cast dropouts under modern strict ANSI validation engines.

Production-Grade Medallion Data Architecture

- **Bronze Layer (Raw Ingestion):** Decoupled schema-on-read layer utilizing dynamic cluster parameters. Preserves original immutable records while attaching Unity Catalog operational lineage (`_metadata.file_path`, `ingested_at`).
- **Silver Layer (Cleaning & Harmonization):** Dynamic schema evaluation mapping variable strings, normalizing split names, running defensive text replacements (`regexp_replace`), and utilizing `try_cast` expressions to safely route malformed text to null values.
- **Gold Layer (Analytical Modeling):** A finalized, business-ready Kimball Star Schema.

Unified Dimensions via SHA-256 Hashing

- **The Collision Vulnerability:** Independent clients frequently issue overlapping native integer keys (e.g., Client A and Client C both reporting `customer_id = 1`).
- **The Structural Solution:** Implemented deterministic, business-key cryptographic hashing via Spark SQL's `try_cast` and `sha2` functions.
- **Surrogate Generation Formula:** `SHA-256(source_client || '|' || client_cust_id)`
- **Architectural Advantages:**
 - Infinite Horizontal Scale: Keys are calculated deterministically across distributed worker nodes in parallel, eliminating the need for relational identity-blocking auto-increments.
 - Perfect Extensibility: New clients (Client D, E, F) can be integrated with zero risk of key collisions or data corruption.

Canonical Analytical Model (Star Schema)

- **dim_customers** (The Universal Dimension Table):
 - Unique Grain: customer_sk (Primary Key)
 - Contains harmonized first names, last names, corporate emails, origin tags, and auditing tracking fields.
- **fct_transactions** (The Event Fact Table):
 - Grain: transaction_sk (Primary Key)
 - Connects to dimension entities via customer_sk (Foreign Key Bridge).
 - Normalizes granular amounts to double-precision values and ensures temporal tracking, utilizing system ingestion timestamps when sources fail to supply raw timestamps.

Defensive Pipeline Engineering in Spark

- **ANSI-Safe Type Casting:** Swapped standard strict Spark casting for SQL `try_cast` expressions. Ensures malformed string anomalies like 'EXTRA' degrade to database NULL rather than triggering fatal job terminations.
- **Data Quality Filters:** Implemented a non-null gating evaluation (`amount.isNotNull()` & `client_txn_id.isNotNull()`) at the edge of the Silver layer to cleanly isolate corrupt records.
- **Decoupled Configuration:** Designed metadata-driven mapping maps (`CLIENT_CONFIGS`) allowing file extensions, patterns, and string delimiters to stay separated from execution logic.

Operational Roadmap for Enterprise Scaling

- **Adding New Formats:** Integrating a new client requires simply adding an entry to a centralized configuration JSON metadata repository—no core processing code needs to change.
- **Advanced Schema Evolution:** To move to a truly enterprise standard, the pipeline can be extended using:
 - Delta Lake Expectations / Great Expectations for continuous runtime data quality auditing.
 - Auto Loader (cloudFiles) for real-time streaming file discovery inside Unity Catalog.
 - dbt Core integration to convert the Silver-to-Gold Python logic into fully audited, self-documenting dbt SQL models.